SIMATS Online Programming Lab

C PROGRAMMING

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CO4 - AT2 - CASE STUDY

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QUESTIONS

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4/4 answered

Q1 CO4 - AT2 - CASE STUDY

25 marks

A large enterprise resource planning application is built from multiple source files, including modules for sales, inventory, and finance, and all of these modules must reference a common variable that stores the current fiscal year. The variable is defined once in one source file but must be read and, in some cases, modified by functions compiled separately in other source files. Without proper declaration, each module would either fail to compile or maintain its own inconsistent copy of the fiscal year. The design must ensure a single, consistent value is shared across the entire application. Question: Identify which storage class allows the fiscal-year variable to be accessed across multiple source files, and explain the role of extern in achieving this shared access.

Your Code

```c
#include <stdio.h>
int fiscal_year = 2024;
extern void sales_print();
extern void inventory_update();
void sales_print() {
    printf("Main: Fiscal Year = %d\n", fi
}
void inventory_update() {
```

```
    fiscal_year = 2025;
}
int main() {

    sales_print();
    inventory_update();
    printf("Updated Fiscal Year: %d\n", f:

    return 0;
}
```

Input

```
Main: Fiscal Year = (print the shared value)
```

Output

Run Save